43. Network Observability: Metrics, Logs, Traces

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Phase: Phase 11 — Observability and troubleshooting
Estimated time: 1–2 days
Slug: `43-observability-net`

Learning objectives

- Define golden network signals
- Instrument latency/errors from clients
- Use flow logs conceptually
- Correlate app traces with net failures

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

Signals: reachability, RTT, loss, DNS latency, TLS latency, saturation, error rates.

```
# continuous simple probe
while true; do
  curl -sS -o /dev/null -w '%{http_code} %{time_total}\n' https://example.com ||
echo fail
  sleep 2
done
```

Cloud flow logs / VPC logs help prove whether packets left a NIC/SG.

Common mistakes

- Metrics without alerts
- Only average latency
- No probe from multiple regions

Exercises

Work in `exercises/43-observability-net/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/43-observability-net/` until after an honest attempt.

Self-check

1. Name four golden signals

2. Why percentiles?
3. Flow logs help prove what?

Further reading

- https://sre.google/sre-book/monitoring-distributed-systems/